differentiate between essential needs and discretionary expenses, and that real people are not as blasé about meeting their “wants” (as opposed to needs) as safety-first advocates assume.

What is the investment approach?

The probability-based approach is based closely on the concepts of maximizing risk-adjusted returns from the perspective of the total portfolio. Asset allocation during retirement is generally defined in the same way during the accumulation phase—using modern portfolio theory (MPT) to identify a portfolio on the efficient frontier in terms of single-period trade-offs between risk and return. Different volatile asset classes that are not perfectly correlated are combined to create portfolios with lower volatility. The efficient frontier identifies the asset allocation combinations with the highest probability-weighted arithmetic average return (often called “expected return” in finance literature) for an acceptable level of year-by-year volatility (often called “risk”). This is an assets-only analysis, and the investor’s spending needs are not part of the decision calculus for determining asset allocation. In addition, inputs for the efficient frontier are generally estimated from historical data. With MPT, investors aim to maximize wealth by seeking the highest possible returns given their capacity and tolerance for risk over a specific time horizon.

For retirement planning, spending and asset allocation recommendations are based on historical or Monte Carlo simulations of failure rates in order to mitigate the risk of wealth depletion inherent in drawing down a portfolio of volatile assets. The failure rate is the probability that wealth is depleted before death or before the end of the fixed time horizon, which stands in for a maximum feasible lifespan. Asset allocation decisions are generally guided by what can minimize the failure rate in retirement. Advocates of the probability-based approach take this as license to use more aggressive asset allocations than seen elsewhere (such as the rule of thumb that bond allocation should be equal to one’s age).

Advice from Bengen and subsequent studies is to have a stock allocation between 50 and 75 percent, but as close as possible to the higher end. Probability-based advocates are generally more optimistic about the long-